11AG-MATH-FINAL — MARKING SCHEME

Teacher copy — not for distribution. Grade 11AG · 26/10/2026 · Total 100 marks

Section A — Easy (6 questions × 4 marks = 24 marks)

Answer all six questions.

1. Find $\lim_{x \to 2} (3x^2 - 5)$. (4 marks)

Answer: 7

The expression is a polynomial, so it is continuous and the limit is the value: $3(2)^2 - 5 = 12 - 5 = 7$.

2. Find $\lim_{x \to 3} \dfrac{x^2 - 9}{x - 3}$. (4 marks)

Answer: 6

Substitution gives $\frac{0}{0}$, so factorise: $\dfrac{x^2 - 9}{x - 3} = \dfrac{(x - 3)(x + 3)}{x - 3} = x + 3$ for $x \neq 3$. As $x \to 3$ this tends to 6.

3. Find $\lim_{x \to \infty} \dfrac{5x^2 + 1}{2x^2 - x}$. (4 marks)

Answer: $\dfrac{5}{2}$

Divide every term by the highest power, x^2: $\dfrac{5 + 1/x^2}{2 - 1/x}$. As $x \to \infty$ both $1/x^2$ and $1/x$ tend to 0, leaving $\dfrac{5}{2}$.

4. Differentiate $y = 4x^5$. (4 marks)

Answer: $20x^4$

Power rule: multiply by the index and drop it by one, $4 \times 5x^4 = 20x^4$.

5. Differentiate $y = \sin x + e^x$. (4 marks)

Answer: $\cos x + e^x$

Differentiate term by term: $(\sin x)' = \cos x$ and $(e^x)' = e^x$.

6. Differentiate $y = \ln x - \cos x$. (4 marks)

Answer: $\dfrac{1}{x} + \sin x$

Term by term: $(\ln x)' = \dfrac{1}{x}$ and $(\cos x)' = -\sin x$, so subtracting it gives $+\sin x$.

Section B — Medium (6 questions × 6 marks = 36 marks)

Answer all six questions. Show your working.

7. Find $\lim_{x \to 1} \dfrac{x^2 + 2x - 3}{x - 1}$. (6 marks)

Answer: 4

Substitution gives $\frac{0}{0}$. Factorising, $x^2 + 2x - 3 = (x+3)(x-1)$, so the quotient is $x + 3$ for $x \neq 1$, which tends to 4.

8. Find $\displaystyle\lim_{x \to \infty} \frac{3x^3 - 2x}{6x^3 + 5}$. (6 marks)

Answer: $\dfrac{1}{2}$

Divide through by x^3: $\dfrac{3 - 2/x^2}{6 + 5/x^3}$. Both small terms vanish as $x \to \infty$, leaving $\dfrac{3}{6} = \dfrac{1}{2}$.

9. Differentiate $y = (2x + 1)^5$. (6 marks)

Answer: $10(2x+1)^4$

Chain rule. The outer derivative is $5(2x+1)^4$ and the inner derivative of $2x + 1$ is 2, so $y' = 5(2x+1)^4 \times 2 = 10(2x+1)^4$.

10. Differentiate $y = x^2 e^x$, giving your answer in factorised form. (6 marks)

Answer: $xe^x(x+2)$

Product rule with $u = x^2$, $v = e^x$: $y' = 2x\,e^x + x^2 e^x = e^x(2x + x^2) = xe^x(x+2)$.

11. Differentiate $y = \tan 4x$, and find the gradient of the curve at $x = 0$. (6 marks)

Answer: $4\sec^2 4x$; gradient 4

Chain rule: the outer derivative is $\sec^2 4x$ and the inner is 4, so $y' = 4\sec^2 4x$. At $x = 0$, $\sec 0 = 1$, so the gradient is $4 \times 1 = 4$.

12. Differentiate $y = \ln\!\left(3x^2 + 1\right)$. (6 marks)

Answer: $\dfrac{6x}{3x^2 + 1}$

Use $\dfrac{d}{dx}\ln f(x) = \dfrac{f'(x)}{f(x)}$ with $f = 3x^2 + 1$ and $f' = 6x$. Forgetting the inner derivative and writing $\dfrac{1}{3x^2+1}$ is the common error.

Section C — Hard (3 questions, 40 marks)

Answer all three questions. Show every step of your working.

13. Evaluate the following limits: (13 marks)

(a) $\displaystyle\lim_{x \to 0} \frac{\sqrt{x+4} - 2}{x}$ (6 marks)

Answer: $\dfrac{1}{4}$

Substitution gives $\frac{0}{0}$, and there is a surd, so multiply above and below by the conjugate $\sqrt{x+4} + 2$. The numerator becomes $(x+4) - 4 = x$, which cancels with the denominator and leaves $\dfrac{1}{\sqrt{x+4} + 2}$. As $x \to 0$ this tends to $\dfrac{1}{2+2} = \dfrac{1}{4}$.

(b) $\displaystyle\lim_{x\to\infty}\left(\sqrt{x^2+3x}-x\right)$ (7 marks)

Answer: $\dfrac{3}{2}$

This is the form $\infty-\infty$. Multiply by the conjugate over itself: $\dfrac{(x^2+3x)-x^2}{\sqrt{x^2+3x}+x}=\dfrac{3x}{\sqrt{x^2+3x}+x}$. Dividing above and below by x (positive, so $x=\sqrt{x^2}$) gives $\dfrac{3}{\sqrt{1+3/x}+1}$, which tends to $\dfrac{3}{2}$.

14. Differentiate, simplifying your answer where possible: (13 marks)

(a) $y=\sqrt{x^2+9}$ (6 marks)

Answer: $\dfrac{x}{\sqrt{x^2+9}}$

Write $y=(x^2+9)^{1/2}$. The chain rule gives $y'=\frac{1}{2}(x^2+9)^{-1/2}\times 2x$, and the 2s cancel to leave $\dfrac{x}{\sqrt{x^2+9}}$.

(b) $y=\ln(\cos x)$ (7 marks)

Answer: $-\tan x$

Use $\dfrac{d}{dx}\ln f=\dfrac{f'}{f}$ with $f=\cos x$ and $f'=-\sin x$, giving $\dfrac{-\sin x}{\cos x}=-\tan x$. The minus sign comes from the derivative of the cosine.

15. A curve has equation $y=x^2\sin x$. (14 marks)

(a) Find $\dfrac{dy}{dx}$. (7 marks)

Answer: $2x\sin x+x^2\cos x$

Product rule with $u=x^2$ and $v=\sin x$: $u'v+uv'=2x\sin x+x^2\cos x$.

(b) Find $\dfrac{d^2y}{dx^2}$. (7 marks)

Answer: $2\sin x+4x\cos x-x^2\sin x$

Differentiate each term of $2x\sin x+x^2\cos x$ by the product rule. The first gives $2\sin x+2x\cos x$; the second gives $2x\cos x-x^2\sin x$. Adding, the two $2x\cos x$ terms combine: $2\sin x+4x\cos x-x^2\sin x$.

— End of the marking scheme —